KLEIDSON ALMEIDA SANTOS

Backend Engineer | Distributed Systems & SaaS Architecture

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SUMMARY

Backend Software Engineer with 5+ years of experience specializing in scalable SaaS platforms, distributed systems, and financial technology (FinTech) integrations. Proven track record of designing high-throughput Node.js/TypeScript microservices and optimizing complex PostgreSQL databases. Expert in system design for distributed reliability, enforcing data integrity, idempotency, and atomic transactions across RESTful APIs and real-time processing pipelines. Driven by delivering mission-critical, fault-tolerant cloud infrastructure for enterprise and B2B products. Fluent in English.

EXPERIENCE

Korven Lab | **Lead Backend Engineer** Dec 2025 – Present | Remote

- Architected and scaled the backend cloud infrastructure for a multi-tenant SaaS platform, engineering a fault-tolerant system capable of processing high volumes of concurrent webhook events with robust high availability.
- Implemented distributed systems patterns (Outbox, exponential backoff) for transactional messaging with optimistic locking, guaranteeing zero data loss and preventing race conditions across distributed deployments.
- Led database design and optimization for complex PostgreSQL architectures, executing 54+ versioned schema migrations with zero downtime and enforcing strict Row-Level Security (RLS) policies.
- Engineered financial ledgers for memberships and commissions utilizing atomic operations and SQL functions, completely eliminating data anomalies and double-payouts under high concurrency.
- Integrated third-party payment APIs (Stripe, Asaas, PagBank, Mercado Pago, Pagar.me) into a unified microservice, managing OAuth onboarding, split routing, and automated reconciliation workflows.
- Enforced enterprise-grade security protocols across all data flows, utilizing timing-safe HMAC signature verification, AES-256-GCM encryption for authentication tokens, and strict API rate limiting.

INSS - National Social Security Institute | **Infrastructure & Support Engineer Intern** Nov 2025 – Present | Brazil

- Developed Python and Shell automation scripts for mission-critical infrastructure maintenance, optimizing routine operational tasks and accelerating critical incident resolution times.
- Maintained internal systems availability, backup data integrity, and strict Identity and Access Management (IAM) protocols.

Appen | **QA Automation Engineer** Jan 2024 – Dec 2024 | Remote, Global Team

- Designed and deployed comprehensive end-to-end automated testing pipelines using Selenium and Python, substantially increasing software test coverage and reducing post-release defects.
- Collaborated within Agile/Scrum release cycles with global engineering teams to identify software edge cases and eliminate post-deployment regressions.

Independent Contractor | **Backend Software Engineer** Mar 2019 – Dec 2023 | Remote

- **B2B API Architecture:** Designed and developed resilient RESTful APIs and normalized relational databases from scratch for enterprise clients, executing complex legacy data migrations with absolute data integrity.
- **Third-Party Integrations:** Engineered secure payment flows and webhook processing pipelines for e-commerce platforms, enforcing robust error handling and transactional idempotency.
- **Full-Lifecycle Delivery:** Managed the full Software Development Life Cycle (SDLC) for high-availability web applications, overseeing cloud infrastructure provisioning, CI/CD pipeline automation, and system performance tuning.

SELECTED SAAS PRODUCTS (PRODUCT ENGINEERING)

- **Papufy (papufy.com) - Services Marketplace:** Engineered a real-time marketplace backend (Node.js) utilizing WebSockets for live bidding. Enforced strict data consistency for digital wallets via PostgreSQL and implemented OWASP-compliant security protocols alongside third-party API split payments.
- **2AVendas (2avendas.com) - B2B Sales CRM:** Architected a multi-tenant B2B order routing engine and CRM API, leveraging Supabase and PostgreSQL to manage complex Role-Based Access Control (RBAC) and high-velocity transactional data.

SKILLS

- **Backend & System Design:** Node.js, TypeScript, Express, RESTful APIs, Microservices Architecture, WebSockets, Zod, JWT, OAuth 2.0, Python
- **Database & Storage:** PostgreSQL (schema design, query optimization, RLS, SQL functions, ACID transactions), Supabase, Relational Databases
- **Distributed Systems:** Outbox pattern, idempotency, background job processing, exponential backoff, rate limiting, High Availability (HA)
- **FinTech Integrations:** Stripe, Asaas, Mercado Pago, Pagar.me (PIX, subscription engines, split routing, webhook API reconciliation)
- **Security & Auth:** HMAC signatures, AES-256-GCM encryption, OWASP Top 10 standards, IAM, secure secrets management
- **DevOps & Infrastructure:** Docker, CI/CD (GitHub Actions), Cloud Computing (AWS EC2, S3), Render, Vercel, node:test, Selenium

EDUCATION & LANGUAGES

- **B.S. Information Systems** – Unifacisa (Expected Dec 2026)
- **Certifications:** Generative AI & Prompt Engineering (NVIDIA / Adapta)
- Introduction to the Brazilian Data Protection Law course Personal Data (ENAP)
- **Languages:** English (Fluent), Portuguese (Native), Spanish (Basic)